First come first serve

Prepared by– Bipasha Majumder, lecturer Department of CSE.

What Is FCFS Scheduling?

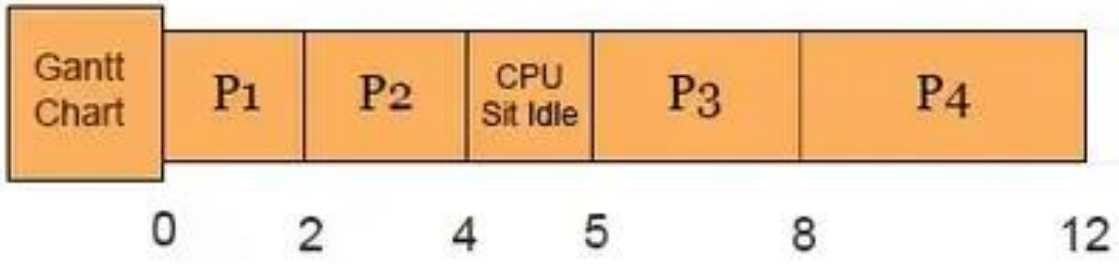
- Simplest CPU scheduling algorithm that schedules according to arrival times of processes.
- The first come first serve scheduling algorithm states that the process that requests the CPU first is allocated the CPU first.
- It is implemented by using the FIFO queue. When a process enters the ready queue, its PCB is linked to the tail of the queue. When the CPU is free, it is allocated to the process at the head of the queue. The running process is then removed from the queue. FCFS is a non-preemptive scheduling algorithm.

- Example-

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time	Response Time	Avg. Time Calculations
P1	0	2					
P2	1	2					
P3	5	3					
P4	6	4					

➢ It Depends on the Arrival Time of the process

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time	Response Time	Avg. Time Calculations
P1	0	2	2	2	0	0	Average Waiting Time = (2+3+3+6)/4 = 3.5
P2	1	2	4	3	1	1	Average Turnaround Time = (0+1+0+2)/4 = 0.75
P3	5	3	8	3	0	0	
P4	6	4	12	6	2	2	



Benefits of FCFS Scheduling

- It is simple and easy to understand.
- FCFS provides fairness by treating all processes equally and giving them an equal opportunity to run.
- FCFS guarantees that every process will eventually get a chance to execute, as long as the system has enough resources to handle all the processes.
- FCFS has low scheduling overhead since it does not involve frequent context switches or complex scheduling decisions.
- FCFS is well-suited for long-running processes or workloads that do not have strict time constraints.

Limitations of FCFS Scheduling

- The process with less execution time suffers i.e. waiting time is often quite long.
- Favors CPU Bound process then I/O bound process.
- Here, the first process will get the CPU first, other processes can get the CPU only after the current process has finished its execution. Now, suppose the first process has a large burst time, and other processes have less burst time, then the processes will have to wait more unnecessarily, this will result in *more average waiting time* .
- This effect results in lower CPU and device utilization.
- FCFS algorithm is particularly troublesome for multiprogramming systems, where it is important that each user get a share of the CPU at regular intervals.

Home work-

Process	Arrival time	Burst time
P1	2	1
P2	3	5
P3	0	2
P4	2	7
P5	4	5

Thank You